RFSoC Mixer Design Notes

Overview of the RF System-on-Chip mixer block used in the high-frequency optical front-end. This document covers signal chain topology, filter specifications, and simulation harness configuration.

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Signal Chain Topology

The full signal path from antenna input to optical modulator output consists of three sequential processing stages:

> **Simby:** [Antenna Input] ——► [LNA Block] ——► [RFSoC Mixer] ——► [IF Filter] ——► [Optical Modulator]

Key design constraints:

- < 3 dB across the full 2–18 GHz sweep band
- +12 dB nominal at 10 GHz centre frequency
- > +25 dBm to handle adjacent channel interferers
- < −60 dBc at the RF port

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Filter Block Specifications

The IF filter following the mixer is a 7th-order Chebyshev bandpass design.

Parameter	Min	Nominal	Max	Unit
Centre frequency	990	1000	1010	MHz
3 dB bandwidth	48	50	52	MHz
Passband ripple	—	0.1	0.5	dB
Stopband rejection	45	—	—	dB
Group delay var.	—	—	2.5	ns

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Workspace Layout

Interactive workspace canvas for the signal chain inspector tool:


```
{
  "view": "signal-chain",
  "blocks": [
    { "id": "lna",      "label": "LNA",               "gain_db": 14 },
    { "id": "mixer",    "label": "RFSoC Mixer",        "lo_freq_ghz": 9 },
    { "id": "if_filt",  "label": "IF Filter",          "bw_mhz": 50 },
    { "id": "opt_mod",  "label": "Optical Modulator","vpi_v": 3.5 }
  ],
  "connections": ["lna->mixer", "mixer->if_filt", "if_filt->opt_mod"]
}
```


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HDL Simulation Snippet

Reference testbench excerpt for the mixer block functional verification:

```
-- RFSoC Mixer Testbench (excerpt)
signal rf_in  : std_logic_vector(11 downto 0);
signal lo_clk : std_logic := '0';
signal if_out : std_logic_vector(11 downto 0);

lo_clk <= not lo_clk after 55.5 ps;  -- 9 GHz LO

uut: entity work.rfsoc_mixer
    port map (
        rf_in  => rf_in,
        lo_clk => lo_clk,
        if_out => if_out
    );
```

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Known Issues

- — the mixer exhibits a transient LO leakage

spike of −40 dBc for the first 500 μs after power-on. Mitigation: hold RF path in blanking state during the warm-up window.

- — resolved in revision 2 via balanced topology.
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References

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- , 4th ed., Wiley, 2012.
- Internal design review slides: doc/reviews/rfsoc_mixer_rev3_review.pdf